Be aware that the maximum credit balance is 2.1 TiB for filesystems smaller than 1 TiB and 2.1 TiB per TiB stored for filesystems larger than 1 TiB. This means filesystems can burst for no more than 12 hours.

It is important to mention that a discount exists for reading data compared to writing data. A filesystem allows 1.66 or three times more read throughput depending on the region. For US East (N. Virginia), the discount is 1.66, which means the maximum read throughput is 5 GiBps. However, be aware that the discount for reading data is not valid for requests returning less than 4 KB of data.

To make things even more complicated, it is worth mentioning that data stored in the Infrequent Access storage class is not taken into consideration when calculating the maximum baseline and burst throughput.

When using the Bursting Throughput mode, you should monitor the burst credits of the filesystem because if the filesystem runs out of burst credits, the throughput will decrease significantly. We've seen many outages caused by an unexpected drop in the baseline performance. The following listing shows how to create a CloudWatch alarm to monitor the `BurstCreditBalance` metric. We've configured the threshold of the alarm to a burst credit of 192 GB, which will allow the filesystem to burst at 100 MiBps for an hour.

Listing 9.7 Monitoring the `PercentIOLimit` metric using a CloudWatch alarm

```
BurstCreditBalanceTooLowAlarm:
  Type: 'AWS::CloudWatch::Alarm'
  Properties:
    AlarmDescription: 'Average burst credit balance over last ...'
    Namespace: 'AWS/EFS'
```